

Huynh Le Dang Khoa

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Summary

Honors B.Eng. Computer Science (HCMUT, GPA 3.6/4.0; completed, diploma pending). Research interests in **computer vision**, **reinforcement learning**, and **vision-language models**, with applications to perception, robot learning, and embodied agents. Contribute to collaborative research at **ARENA Lab** (SMU) and **SMART** on segmentation, simulation, and world models.

- Co-authored **HORSE** (AAAI'27 under review): geometry-grounded relational reranking for referring segmentation on frozen **SAM3** with CLIP text.
- Experience with **PyTorch**, 2D/3D multimodal perception, Isaac Lab / ROS 2, video / latent **world models**, and GPU-efficient inference systems.

Education

Ho Chi Minh City University of Technology (HCMUT – VNU-HCM) Ho Chi Minh City, Vietnam
B.Eng. Computer Science & Engineering — Honors Program (Top 5% of Faculty) 2022 – 2026

- **GPA:** 3.6 / 4.0 · [Academic Transcript](#)
- **Focus:** Machine learning, computer vision, deep learning, and systems for ML inference.

Experience

Eye Code Tech — ARENA Lab @ Singapore Management University **Singapore (Remote)**
Research Collaborator — Vision AI, Simulation & Robot Learning Feb 2026 – Present

- **Collaborative Research:** Contribute to joint projects at **ARENA Lab** (Singapore Management University; Berlin & Singapore) and **SMART** (Singapore-MIT Alliance for Research and Technology) on vision, simulation, and robot learning.
- **HORSE (VLM-grounded Segmentation):** Co-authored **HORSE**: geometry-grounded relational reranking for **SAM3** referring segmentation with CLIP text (AAAI'27 under review). Train and ablate at scale on **8× NVIDIA H100**; [OpenReview].
- **Multi-Modal Perception:** Train and evaluate **object detection**, 6-DoF pose, depth, and segmentation models in **PyTorch**; design ID/OOD and sim-to-real benchmarks combining **2D**, **3D**, and scene context.
- **World Models & Robot Learning:** Experiment with latent / video-style **world models** for predictive scene understanding and synthetic trajectories; contribute to **Arena-RosNav** navigation in Gazebo/**Isaac Lab** with **ROS 2**; [GitHub].

- **High-Performance Inference (Thesis):** Led adaptive scheduling for mixed ML workloads: packaged **YOLO/ResNet**, profiled **latency**/throughput/memory on **NVIDIA RTX 5090**, and tuned batching under SLA targets for real-time serving.
- **Systems & Evaluation:** Designed reproducible benchmarks vs. strong baselines; analyzed accuracy–latency trade-offs for GPU-accelerated inference pipelines and documented results for peer review.

Projects

HORSE — AAAI 2027 Submission (Co-Author) · Lightweight **geometry-grounded relational reranking** for referring expression segmentation on frozen **SAM3** with CLIP — language-conditioned candidate selection without mask-generator retraining; [OpenReview].

- ▷ *Method:* Box geometry + self-attention + frozen CLIP text; fuse with backbone confidence — no mask-generator retraining.
- ▷ *Results:* Improves SAM3 on all nine RefCOCO-family splits (e.g. RefCOCO+ TestB **+10.96**); strong on relation-intensive queries.
- ▷ *Stack:* Python, **PyTorch**, SAM3, CLIP.

Video World Models for Robot Generalization — with Do Van Minh; Inspired by DreamGen · Exploring **DreamGen**-style pipelines that use video world models to synthesize neural trajectories for robot policy learning — generating diverse behaviors/environments from limited teleoperation data, then recovering pseudo-actions for training; [arXiv:2505.12705].

- ▷ *Focus:* Image-to-video world models adapted to robot embodiments; latent-action / inverse-dynamics recovery; scaling beyond manual data collection for behavior and environment generalization.
- ▷ *Stack:* Python, **PyTorch**, video generative models, Isaac Lab / robot simulation, policy learning evaluation.

Adaptive ML Inference Server — Graduation Thesis, Score 9.84/10 (Highest in Cohort) · Adaptive scheduling for concurrent inference under bursty traffic; packaged **YOLO/ResNet**, profiled on **RTX 5090**; ~**20%** higher throughput and ~**80%** fewer SLA violations vs. A* baselines.

- ▷ *Stack:* Python, **ONNX Runtime**, GPU latency & memory profiling, Linux.

Publications & Achievements

Publications: *HORSE: Geometry-Grounded Relational Reranking for Referring Expression Segmentation.* Co-Author, AAAI 2027 (under review). [OpenReview](#) · *An Intelligent Virtual Machine Consolidation Approach...* First Author, 19th ACOMPA (2025). [DOI](#)

Skills: **Python**, **C++**; Linux; **PyTorch**, OpenCV, CUDA; CLIP / VLM-style text–vision fusion; SAM3, YOLO, ResNet; **Isaac Lab**; ROS 2; ONNX packaging; Git.

Awards: Most Outstanding Intern — AI Engineer, Dien Toan Group; 2nd Place, Innostart Entrepreneurship Competition (AI image-generative model startup); 4th Place, UTE Junior Startup.

Languages: English (Professional Working Proficiency); Vietnamese (Native).